

HARISH VELAYUTHAM

Vilnius, Lithuania | +370 65425110 | harishmech415@gmail.com

Portfolio: <https://harish-ai-automation-portfolio.vercel.app/>

PROFILE

AI Automation Engineer with 2+ years of practical experience designing and implementing AI-powered workflows, API integrations and business process automation systems. Specialised in n8n orchestration, JavaScript/Node.js, Python and integrating LLMs into modular automation pipelines. Experienced in building multi-agent workflows, Retrieval-Augmented Generation (RAG) systems, document-processing pipelines, and automated content workflows using OpenAI, Claude, Gemini, Ollama, and Supabase. Strong focus on reliable integrations, structured data processing, error handling and maintainable automation architecture.

EDUCATION

Kaunas University of Technology, Lithuania

Masters in Industrial Engineering & Management | 2022 - 2024

Sri Ramakrishna Institute of Technology

Bachelor's in Mechanical Engineering | 2018 - 2021

PROFESSIONAL EXPERIENCE

- **AI Automation Specialist — Freelance**
January 2025 – Present | Lithuania
- **Head of AI and Automation(Freelance contract) — VOILA IA**
December 2025 – April 2026 | Nice, France
- **AI Automation Service — Independent Contract, UNIS Company Inc.**
April 2025 – June 2025 | Walnut, California, USA
- **AI Automation Projects Manager — Cookiejar AI**
October 2024 – November 2024 | Kaunas, Lithuania
- **AI Automation Assistant — Cookiejar AI (Internship)**
June 2024 – September 2024 | Kaunas, Lithuania

TECHNICAL SKILLS

Workflow Automation: n8n Cloud and self-hosted, Make.com, Zapier, webhooks, event-driven workflows, scheduled automation, API orchestration.

Programming & Backend: JavaScript, Node.js, Python, TypeScript, FastAPI, Flask, REST APIs, JSON, GraphQL.

AI & LLM Integration: OpenAI APIs, GPT models, Claude, Gemini, Ollama, Jev AI, LangChain, multi-agent workflows, prompt engineering, AI-assisted development with OpenAI Codex.

RAG & Data Processing: Retrieval-Augmented Generation, embeddings, Supabase Vector, PostgreSQL, semantic search, document processing, transcript ingestion, OCR, Tesseract, EasyOCR.

Integrations & Business Systems: Discord API, YouTube API, Google Sheets, Airtable, SMTP/email automation, HubSpot, Pipedrive, Zoho, OAuth 2.0.

Media & Content Automation: FFmpeg, audio/video processing, Whisper, transcript extraction, ComfyUI, AI image generation, AWS S3.

Infrastructure & Reliability: Docker, Linux, Hostinger VPS, environment configuration, logging, error handling, workflow monitoring and failure recovery.

1. Discord AI Assistant — RAG & Knowledge Automation

Technologies: n8n Cloud, Discord API, Supabase, OpenAI embeddings, YouTube transcripts, LLMs

- Developing a knowledge-grounded AI assistant for educational content, enabling students to ask questions based on course materials and teacher-specific knowledge.
- Designing a retrieval pipeline using Supabase vector storage and OpenAI embeddings to retrieve relevant information from course transcripts.
- Building n8n workflows to process YouTube transcript data and maintain structured knowledge sources for retrieval.
- Designing teacher-specific routing and assistant personas to support distinct course knowledge bases and response behaviour.
- Implemented short-term conversational memory to preserve recent context across student interactions.

2. Automated Discord Content Processing System — OCR & AI Redaction

Technologies: Node.js, REST APIs, Tesseract, EasyOCR, LLMs, Discord API

- Built an end-to-end pipeline to process payout images, extract structured information and publish formatted updates to Discord.
- Developed a custom Node.js backend to coordinate OCR extraction, data parsing, validation and workflow execution.
- Implemented coordinate-based masking to redact sensitive information from images before publication.
- Integrated AI-assisted extraction and validation to improve consistency in structured payout data.
- Automated the processing and publishing workflow, reducing manual handling and moderation effort.

3. Modular Multi-Agent AI Workflow Chatbot

Technologies: n8n, Gemini, Claude, OpenAI, Supabase

- Built a modular conversational workflow using dedicated sub-workflows for response generation, evaluation and synthesis.
- Used Gemini to generate initial responses, Claude to critique response quality, and OpenAI to synthesise final answers.
- Integrated Supabase-backed session memory and file context to support context-aware responses.
- Designed modular workflow components to simplify maintenance and independent model updates.

4. Invoice AI Processing System

Technologies: AI document processing, OCR, Airtable, Google Sheets, Slack

- Developed an AI-assisted invoice processing workflow for PDFs, DOCX documents and images.
- Extracted and mapped invoice information into predefined Airtable or Google Sheets fields.
- Implemented missing-field checks and Slack notifications to flag incomplete invoice data.

5. Automated Media Generation for Glossary Entries

Technologies: n8n, GraphQL, ComfyUI, LangChain, Supabase, AWS S3

- Developed an automated pipeline to retrieve glossary entries from a headless CMS and generate relevant illustrative images.
- Integrated AI-based prompt generation and ComfyUI image generation into the workflow.
- Uploaded generated assets to AWS S3 and updated CMS entries using GraphQL mutations.
- Implemented batch processing, prompt sanitization and Supabase-based tracking to manage execution and prevent unnecessary reprocessing.

6. Daily YouTube Digest Automation

Technologies: n8n, YouTube API, Kome AI, OpenAI, LangChain, SMTP

- Automated retrieval of new videos from selected YouTube channels and extraction of video transcripts.
- Integrated LLM-based summarisation with terminology normalisation to produce technical summaries.
- Generated and emailed branded HTML digests containing video thumbnails, links and summaries.
- Implemented scheduled execution to automate recurring content collection and distribution.

7. AI Product Recommendation Assistant

Technologies: n8n, LLM APIs, CRM integrations, email automation

- Designed an automated email-response workflow for product inquiries and quotation requests.
- Integrated CRM data to distinguish existing customers from new prospects and support personalised recommendations.
- Implemented product-link analysis and matching logic to identify relevant products for customer requests.
- Automated response preparation to reduce manual handling and improve consistency.

8. Job Posting Automation System

Technologies: n8n, LLM APIs, web integrations, custom forms

- Developed workflows to generate job descriptions, tracking-enabled URLs and recruitment posters.
- Automated distribution of job advertisements across supported platforms.
- Created a template-driven poster generation workflow with user input through a custom interface.